

Anuj Pal

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SUMMARY

Computer Science sophomore, research intern, and NCAA D-III varsity athlete. Experienced in building full-stack web applications, REST APIs, and real-time data pipelines that turn live sensor and hardware data into usable software. Seeking a **Summer 2027 software engineering internship**.

EDUCATION

University of Wisconsin-Whitewater

Fall 2026 – May 2029 (Expected)

Bachelor of Science in Computer Science

Whitewater, WI

- **Relevant Coursework:** Data Structures, Algorithms, Python Programming, Calculus, Java

Luther College

Aug. 2025 – May 2026

Coursework toward B.A. in Computer Science (transferred)

Decorah, IA

EXPERIENCE

Research Intern, Embedded Systems & Biomechanics

Jun. 2026 – Aug. 2026

Indian Institute of Technology (IIT) Delhi & IIT Roorkee

India

- Built a real-time impact-testing system (**Python, Flask, Socket.IO**) at IIT Delhi's Centre for Biomedical Engineering (CBME), streaming live sensor readings over WebSockets to a live **Chart.js** dashboard (github.com/palan01-droid/Drop_tower)
- Designed and wired a **multi-layer FSR 402 sensor circuit** modeling skin, skull, and brain layers to measure how impact force transfers through a physical head model, going from sensor and circuit fundamentals to a working data-acquisition pipeline within weeks in a structured research environment with regular progress reviews
- At IIT Roorkee's Extreme Mechanics Laboratory (Jun. – Jul. 2026), engineered an **IoT crash-detection device** for automotive safety under Prof. Shailesh Ganpule, building Arduino sensor circuits and Python pipelines for real-time impact-data collection, and built a live visualization GUI integrated with a companion mobile application

PROJECTS

Sensor Monitor | *Python (Flask, Socket.IO), Supabase, Groq API, Docker*

Live Demo · GitHub

- Built a full-stack IoT dashboard supporting live Arduino/ESP32 hardware (DHT11, MPU-6050, HC-SR04, MQ-2/MQ-3, LDR, vibration, IR sensors), a hardware-free simulation mode, and an HTTP ingest endpoint for WiFi-connected boards
- Integrated the **Groq API (Llama 3.3 70B)** to power a conversational assistant that answers natural-language questions about live sensor data and auto-explains threshold-breach alerts
- Implemented Google OAuth via Supabase and SQLite/PostgreSQL storage, backed by **23 pytest tests** covering auth, ingestion, and input validation; containerized with Docker and deployed live on Render

AI Study Assistant (RAG Pipeline) | *Python (Flask, ChromaDB, Ollama)*

GitHub

- Built a local retrieval-augmented generation pipeline that parses course PowerPoint decks with **python-pptx**, embeds and indexes slide content in a persistent **ChromaDB** vector store, and grounds answers using a locally-hosted **Llama 3.2** model via Ollama
- Designed a query-intent router that switches between explanation and quiz-generation prompts; initially prototyped with LangGraph, then replaced with a lightweight keyword-based if/else after concluding the added complexity wasn't justified for a two-way routing decision

CERTIFICATIONS

Software Engineering Job Simulation | JPMorganChase (Forage)

Jul. 2026

- Integrated Kafka into a Spring Boot microservice to consume and deserialize transaction messages, and implemented persistence logic with Spring Data JPA and an H2 database
- Built and tested REST endpoints (Spring, Maven) for user balance queries and integration with an external incentive API

TECHNICAL SKILLS

Languages: Python, Java, SQL

Frameworks & Libraries: Flask, Socket.IO, Chart.js, ChromaDB

Hardware & Embedded: Arduino, ESP32, Circuit Design, FSR Sensors

AI & Data: LLM Integration (Groq, Ollama), Vector Search (ChromaDB), Real-Time Data Pipelines

Tools & Platforms: Git, GitHub, Supabase, Render, Vercel, REST APIs, AWS

ACTIVITIES & AWARDS

Men's Varsity Tennis, Luther College: American Rivers Conference Champions (2024–25); balanced NCAA D-III athletics with full-time CS coursework

Bronze Medal, SGFI National School Games: Led Chandigarh nationals tennis team U-19; Ranchi, India (2024–25)